

SYNOPSIS Presentation
on
Smart Campus Management
Portal

Manish Kumar Singh (202410116100114)

Mansi Garg (202410116100116)

Mukul Dhiman (202410116100126)

Session:2025-2026 (IV Semester)

Under the supervision of

Dr. Ankit Verma (Associate Professor)

Department Of Computer Applications

KIET GROUP OF INSTITUTIONS, DELHI-NCR, GHAZIABAD
201206





Project Overview

A comprehensive web-based system designed to digitally manage all college activities, eliminating manual record-keeping and communication gaps.

Unified Portal

Single integrated platform for students, teachers, and administrators

Core Features

Attendance tracking, assignment management, results publication, and notice distribution

Impact

Transforms campus operations into a paperless, organized, and efficient ecosystem

Project Objectives



1 Digital Centralization

Create a unified system that maintains all student and academic records digitally

2 Paperless Operations

Reduce paperwork by enabling teachers to upload attendance, marks, and assignments online

3 Enhanced Communication

Enable seamless access to notices, results, and updates for students

4 OBE Support

Align with Outcome-Based Education principles for measurable learning outcomes

Project Scope



Admin Module

- Add and manage student records
- Add and manage teacher profiles
- Course and curriculum management
- Upload campus-wide notices



Teacher Module

- Mark and track student attendance
- Upload and manage assignments
- Enter and publish marks
- Monitor student performance



Student Module

- View attendance records
- Submit assignments online
- Access exam results
- Read important notices

📄 Future Enhancements: Online fee payment, hostel management, and transport tracking will be added in subsequent phases.

Methodology & Technologies

Development Approach

We adopted **Agile methodology** to develop the system iteratively in small, manageable modules. This approach allows for continuous testing, feedback integration, and flexible adaptation to changing requirements.

Technology Stack

Built on the robust **MERN Stack**, our portal leverages modern web technologies for optimal performance and scalability.

Frontend

React.js - Dynamic, responsive user interface with component-based architecture

Backend

Node.js with Express.js - Fast, scalable server-side application framework

Database

MongoDB - Flexible NoSQL database for efficient data management

Development Tools

VS Code, GitHub, Postman, Chrome/Edge Browser

Hardware & Software Requirements

Hardware Specifications

- Computer or Laptop
- Minimum 4GB RAM
- Stable Internet Connection
- Adequate storage space (20GB+)

Software Environment

- Operating System: Windows 10/11 or macOS
- Web Browser: Chrome or Microsoft Edge
- VS Code (Code Editor)
- Node.js Runtime Environment
- MongoDB Database System
- Postman (API Testing)

Project Timeline

An 8-week structured development plan ensuring systematic progression from concept to deployment.



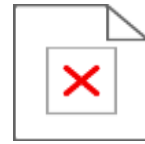
SUCSESUJIL WEB WEB APPLICIANCTN

www.visstudies. A6b doakic ka oon



Project Outcomes

A fully functional web application delivering tangible benefits to all stakeholders in the campus ecosystem.



Online Attendance

Real-time attendance tracking system accessible to students and faculty



Digital Results

Instant result publication and performance tracking for students



Notice Board

Centralized announcement system ensuring no one misses important updates

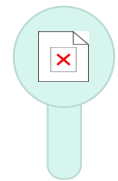


Time & Resource Savings

Significant reduction in paperwork and administrative overhead

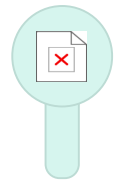
Future Enhancements

Expanding capabilities to create a comprehensive, next-generation campus ecosystem.



Mobile Application

Native iOS and Android apps for on-the-go access



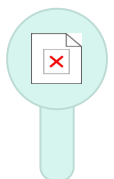
Online Fee Payment

Integrated payment gateway for tuition and other fees



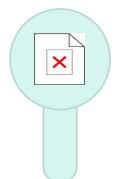
SMS & Email Alerts

Real-time notifications for important updates



AI Performance Analysis

Predictive analytics for student performance insights



Face Recognition

Biometric attendance marking for enhanced accuracy

Vision for Tomorrow

These enhancements will transform our portal into a comprehensive smart campus solution, leveraging cutting-edge technologies like AI and biometrics to create an unparalleled educational experience.